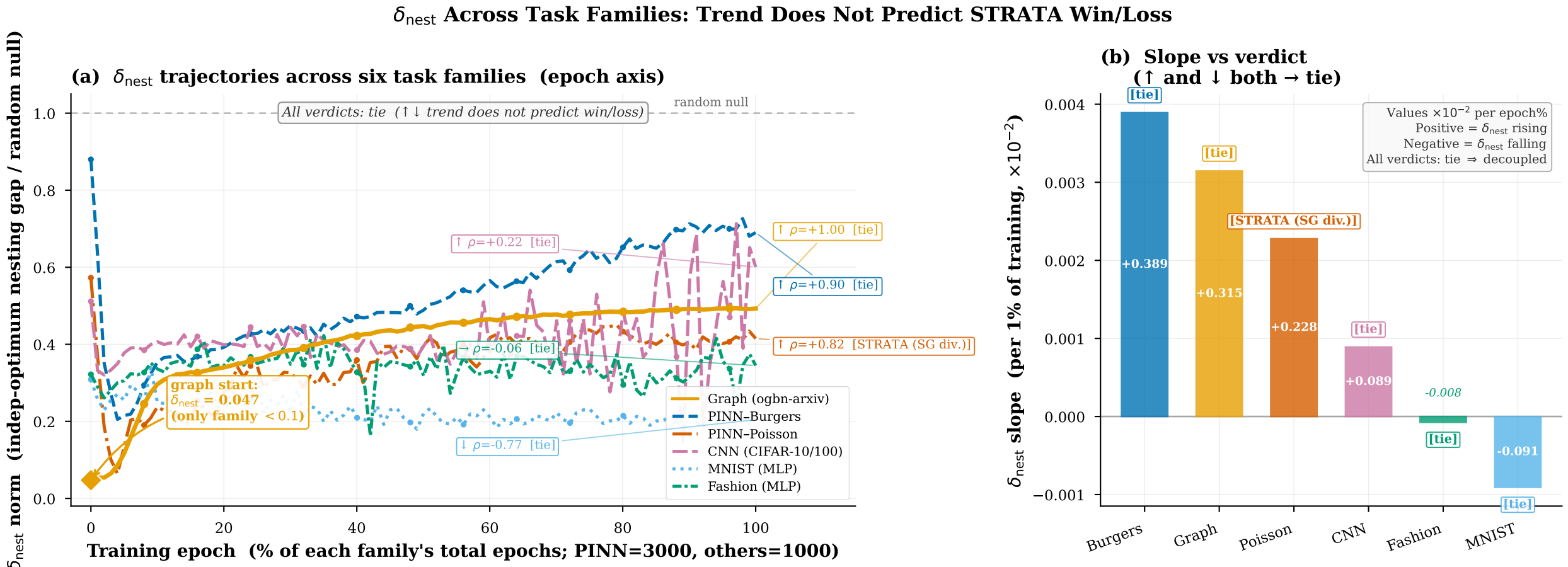

δnest Across Task Families: Trend Does Not Predict STRATA Win/Loss
(a) δnest trajectories across six task families (epoch axis)
All verdicts: tie (↑↓ trend does not predict win/loss)
random null
↑ ρ=+0.22 [tie]
↑ ρ=+1.00 [tie]
↑ ρ=+0.90 [tie]
↑ ρ=+0.82 [STRATA (SG div.)]
→ ρ=-0.06 [tie]
↓ ρ=-0.77 [tie]
graph start:
δnest = 0.047
(only family < 0.1)
Graph (ogbn-arxiv)
PINN–Burgers
PINN–Poisson
CNN (CIFAR-10/100)
MNIST (MLP)
Fashion (MLP)
δnest norm (indep-optimum nesting gap / random null)
Training epoch (% of each family's total epochs; PINN=3000, others=1000)
(b) Slope vs verdict
(↑ and ↓ both → tie)
Values ×10⁻² per epoch%
Positive = δnest rising
Negative = δnest falling
All verdicts: tie ⇒ decoupled
δnest slope (per 1% of training, ×10⁻²)
[tie]
+0.389
[tie]
+0.315
[STRATA (SG div.)]
+0.228
[tie]
+0.089
-0.008
[tie]
-0.091
[tie]
Burgers
Graph
Poisson
CNN
Fashion
MNIST